

when working with stochastic volatility what is the metrics using when comparing the already extracted latent variables in options such as IBM, for exemple lets say a want to compare two points in time using stochastic volatility and I have extracted both thoses periods Latent Volatility variables such as Sigma, Phi and Mu what are the academic prosedure to comparing these variables against each other

Comparing Latent Parameters in Stochastic Volatility Models: Academic Approaches

When working with stochastic volatility (SV) models, researchers typically treat μ , ϕ , σ and the latent volatilities as *random variables with posterior distributions*, not single point estimates. Comparisons across time or subsamples are therefore usually done via **Bayesian posterior comparison** or **model comparison**, rather than ad-hoc metrics.

How Parameters and Latent Volatility Are Estimated

- SV parameters and latent volatilities are usually estimated with **Bayesian methods**: MCMC, particle MCMC, INLA, or approximate Bayesian computation, yielding full posterior draws for μ , ϕ , σ and h_t (log-volatility) (Kastner, 2016; Chaim & Laurini, 2024; Vankov et al., 2019; Shapovalova, 2021; Hosszejni & Kastner, 2019; Omori et al., 2007).
- For IBM options and prices, latent volatility and persistence parameters are estimated with instrumental-variable methods and robust confidence sets for the persistence parameter (analogous to ϕ) (Ahsan & Dufour, 2025).

Comparing Parameters Across Time / Samples

Posterior-based comparison

With posterior draws for each period or model:

- **Credible intervals / overlap**: compare 95% posterior intervals for μ , ϕ , σ between two periods; non-overlap suggests meaningful change (Chaim & Laurini, 2024; Shapovalova, 2021; Omori et al., 2007).
- **Posterior probabilities of ordering**: compute $P(\phi_1 > \phi_2 \mid \text{data})$ or similar from MCMC draws; used implicitly in simulation studies that assess recovery and uncertainty of μ , ϕ , σ (Vankov et al., 2019; Shapovalova, 2021; Omori et al., 2007).
- **Posterior summaries of latent volatility paths** (means, quantiles) and visual comparison of smoothed volatilities over time (Yu & Meyer, 2006).

Model-comparison criteria

When comparing SV specifications (or the same SV model fit on different subsamples):

- **Marginal likelihoods / Bayes factors** and **Deviance Information Criterion (DIC)** are standard (Chan, 2022; Chan & Eisenstat, 2018; Yu & Meyer, 2006; Gerlach & Tuyl, 2005).
- These integrate out the latent volatilities and parameters, providing a scalar measure to rank models/periods given μ , ϕ , σ differences (Chan, 2022; Chan & Eisenstat, 2018; Yu & Meyer, 2006; Gerlach & Tuyl, 2005).

Example metrics used in the literature

Goal	Typical metric / tool	Citations
Rank SV specifications or regimes	Marginal likelihood, Bayes factor, DIC	(Chan, 2022; Chan & Eisenstat, 2018; Yu & Meyer, 2006; Gerlach & Tuyl, 2005)
Assess parameter change (μ, ϕ, σ)	Posterior means, credible intervals, ESS diagnostics	(Kastner, 2016; Chaim & Laurini, 2024; Vankov et al., 2019; Shapovalova, 2021; Hosszejni & Kastner, 2019; Omori et al., 2007)
Compare latent volatility paths	Smoothed volatility plots, RMSE vs “true” in simulations	(Vankov et al., 2019; Yu & Meyer, 2006)
Compare fit/forecast performance	Log predictive likelihoods, out-of-sample scores	(Chan, 2022; Chaim & Laurini, 2024; Nacinben & Laurini, 2024; Nikolakopoulos, 2025)

FIGURE 1 Common quantitative tools for comparing SV parameters and volatility.

Conclusion

In academic work on stochastic volatility, μ , ϕ , σ and latent volatilities from different time periods are compared using **Bayesian posterior summaries and formal model-comparison criteria**, not simple pointwise differences. Standard practice is to (1) estimate full posteriors for each period/model, (2) compare posterior means and credible intervals or posterior probabilities of parameter differences, and (3) use marginal likelihoods, Bayes factors, DIC, and predictive scores to judge which period/model better explains the data.

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